12	**(a)**		$y=\sin\left(x+\frac{\pi}{2}\right)$ is a translation of $y=\sin x$ $\frac{\pi}{2}$ to left and so is the same as $y=\cos x$	**M1** **E1** **[2]**	**3.1a** **2.2a**	OR $\sin\left(x+\frac{\pi}{2}\right)=\sin x\cos\frac{\pi}{2}+\sin\frac{\pi}{2}\cos x$ OR $\sin\left(x+\frac{\pi}{2}\right)=\sin x\times 0+1\times\cos x=\cos x$ Convincing completion (AG)
12	**(b)**		$[\cos x\approx]\frac{16\left(x+\frac{\pi}{2}\right)\left(\frac{\pi}{2}-x\right)}{5\pi^2-4\left(x+\frac{\pi}{2}\right)\left(\frac{\pi}{2}-x\right)}$ $[\cos x\approx]\frac{16\left(\frac{\pi^2}{4}-x^2\right)}{5\pi^2-4\left(\frac{\pi^2}{4}-x^2\right)}$ $[\cos x\approx]\frac{4\pi^2-16x^2}{4\pi^2+4x^2}=\frac{\pi^2-4x^2}{\pi^2+x^2}$	**M1*** **DM1** **A1** **[3]**	**2.1** **1.1** **1.1**	Substitute $x+\frac{\pi}{2}$ Multiplying out brackets (dep on first M1) Convincing completion (AG) Check continuation page